

TUOTUOHE, China

WMO: 560040

Lat: 34.2161N

Lon: 92.4385E

Elev: 14879

StdP: 8.33

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-12.6	-33.5	1.5	0.3	-30.5	1.7	3.8	41.3	29.5	37.2	24.5	3.1	270	0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	63.4	43.4	60.5	43.0	57.9	42.4	47.5	57.5	46.4	56.1	45.2	54.0	10.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
43.6	74.2	51.5	42.4	70.9	50.2	41.4	68.1	49.3	24.7	57.7	23.9	56.2	23.0	54.4	52.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.5	26.8	22.7	DB	-22.3	68.8	5.8	3.1	-26.5	71.0	-29.9	72.8	-33.1	74.5	-37.4	76.7	(4)
(5)			WB	-23.1	49.7	5.5	1.2	-27.0	50.6	-30.3	51.2	-33.4	51.9	-37.4	52.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	6.0	10.5	17.9	27.0	35.9	43.3	48.1	47.5	41.2	28.4	15.4	8.0	(6)
(7)		DBStd	16.02	6.70	6.25	6.00	5.58	4.48	4.60	3.87	3.94	4.63	6.74	5.87	6.62	(7)
(8)		HDD50	8253	1365	1107	995	690	438	205	82	97	263	669	1039	1303	(8)
(9)		HDD65	13682	1830	1527	1460	1140	903	651	523	544	713	1134	1489	1768	(9)
(10)		CDD50	47	0	0	0	0	0	4	24	19	0	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.5	9.7	10.7	10.7	9.7	9.0	8.0	7.3	6.9	6.6	7.1	8.0	8.6	(14)
(15)	Precipitation	PrecAvg	10.5	0.0	0.0	0.1	0.2	0.6	2.0	3.2	2.5	1.6	0.4	0.0	0.0	(15)
(16)		PrecMax	15.3	0.2	0.2	0.4	0.6	2.2	4.0	6.9	5.4	3.2	3.1	0.2	0.3	(16)
(17)		PrecMin	6.4	0.0	0.0	0.0	0.0	0.0	0.5	0.6	0.3	0.3	0.0	0.0	0.0	(17)
(18)		PrecStd	2.3	0.0	0.0	0.1	0.1	0.5	0.8	1.2	1.2	0.7	0.6	0.0	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.8	45.1	51.4	60.9	65.2	67.5	66.8	59.9	54.5	38.8	35.5	(19)
MCWB			20.7	21.8	26.3	29.8	35.4	41.2	45.2	43.9	42.5	33.6	23.3	21.4	(20)	
2%		DB	28.6	32.0	39.4	47.7	54.9	59.8	63.4	63.2	56.6	47.6	35.0	30.0	(21)	
		MCWB	16.9	19.3	23.6	28.6	33.5	41.8	44.4	43.9	41.3	31.6	21.5	17.8	(22)	
5%		DB	24.1	28.6	36.0	44.2	51.0	56.5	60.5	60.0	54.0	43.4	32.1	26.0	(23)	
		MCWB	14.2	17.2	22.0	27.3	32.3	41.0	43.8	43.6	40.6	29.8	19.9	15.5	(24)	
10%		DB	20.3	25.1	32.7	41.0	47.6	53.3	57.7	57.1	51.0	39.8	28.9	22.1	(25)	
		MCWB	12.0	15.2	20.3	26.1	31.6	40.0	43.3	43.2	39.4	28.0	18.3	13.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	22.3	26.8	31.7	39.5	47.3	49.3	48.6	46.1	36.9	24.4	21.3	(27)
MADB			34.3	35.5	42.6	45.1	49.7	56.8	60.0	58.9	55.3	47.0	36.3	34.7	(28)	
2%		WB	17.1	19.5	24.2	29.8	36.7	45.1	47.9	47.2	44.2	34.0	22.1	18.1	(29)	
		MADB	27.7	31.1	37.6	43.8	47.1	54.8	58.1	57.3	53.4	43.8	33.5	29.3	(30)	
5%		WB	14.5	17.6	22.5	28.4	34.9	43.7	46.7	46.1	42.8	31.8	20.4	15.7	(31)	
		MADB	23.6	27.7	34.7	41.5	45.2	52.7	56.5	55.8	51.4	40.1	31.1	25.8	(32)	
10%		WB	12.1	15.6	20.7	27.0	33.2	42.1	45.5	45.0	41.2	29.6	18.7	13.3	(33)	
		MADB	20.0	24.7	31.9	39.0	43.5	50.0	54.4	53.6	48.3	37.6	28.3	21.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	27.7	27.1	27.1	25.1	21.5	18.5	18.4	18.3	18.7	21.2	27.0	28.4	(35)	
MCDBR			32.0	32.0	31.1	30.5	28.7	23.4	22.9	23.6	22.8	25.1	30.4	33.1	(36)	
MCWBR			22.8	21.9	18.7	15.3	11.7	9.0	8.0	8.3	9.5	11.0	19.1	23.2	(37)	
5% WB		MADB	31.4	30.6	29.2	25.9	22.0	18.4	17.2	17.3	18.3	21.2	29.0	32.3	(38)	
		MCWBR	22.7	21.2	17.9	13.4	10.2	8.5	7.3	7.2	8.4	9.7	18.5	22.8	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.174	0.195	0.237	0.278	0.295	0.290	0.287	0.280	0.252	0.217	0.181	0.168	(40)	
taud		2.624	2.562	2.435	2.323	2.338	2.460	2.535	2.550	2.597	2.626	2.640	2.676	(41)		
Ebn at Noon		347	345	333	320	313	313	313	316	324	332	341	346	(42)		
Edh at Noon		23	27	34	40	40	36	33	32	29	25	22	20	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1065	1308	1655	1931	2110	2063	2082	1915	1704	1470	1229	1050	(44)	
(45)		RadStd	91	104	92	102	96	83	125	105	92	97	50	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.19	+2.93	N/A	N/A	+0.67	N/A	-439	-457	N/A	N/A
(47) Regional (1 neighbor)		+1.01	+2.02	N/A	N/A	+0.65	+0.47	-439	-457	N/A	N/A

Nomenclature: See separate page